

Churn Analysis and Customer Intelligence



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Churn Analysis & Customer Intelligence



1. The Business Challenge

2. In the highly competitive subscription-based market, **customer retention is critical for sustainable business growth.**
3. This project focuses on analyzing customer churn using a multi-dimensional dataset containing **customer demographics, subscription details, financial information, customer satisfaction, complaints, escalations, and churn indicators.**
4. As a Data Analyst, the objective is to identify **high-risk customer segments**, understand the key factors associated with churn, and translate data-driven findings into actionable business insights that can support customer retention strategies.

5. Core Tech Stack

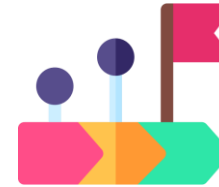
SQL & Python Integration (*numpy, pandas, sqlite3, matplotlib, seaborn*), Performance Engineering (*data cleaning, feature engineering, analytics*), Behavioral Visualization, Writing Actionable Insights

6. Project Milestones

- Relational Data Extraction: Connecting Python to SQL databases to pull multi-table datasets.
- Advanced Feature Engineering: Data imports, calculating tenure, churn rates, and customer aging.
- Executive Reporting: Translating technical findings into billion-dollar business insights.

Note: *This isn't just a coding exercise, it's a business solution. You'll leave with a portfolio-ready project applicable to E-commerce, SaaS, Fintech and Adtech industries.*

Churn Analysis - Roadmap



- Connect SQL database to Python — *pandas and sqlite3*
- Data import using SQL query in Python — *pandas and sqlite3*
- Data Cleaning — *numpy and pandas*
 - data types, rename cols, select specific cols, QCs, handle missing/null values
- Feature Engineering — *numpy and pandas*
 - Create new calculated cols, data transformation, use filters
- Data Analysis — *numpy and pandas*
 - EDA - aggregation, group by, pivot table
- Data Visualization — *matplotlib and seaborn*
- Insights on ppt — *key insights, next steps, impact to show in resume*

Churn Definition by Business



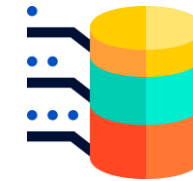
Churn analysis is the process of figuring out why customers stop doing business with you.

For this, we only need to look at three things:

- **"Who"**: Identifying which customers left or will leave in future
- **"Why"**: Analyzing their behavior before they left.
- **"When"**: Finding the "danger zone" – point when a user leave

Business Type	Churn Definition
SaaS	Subscription canceled
E-commerce	No purchase in 90 days
Adtech	Using service/app in 90 – 120 days
Streaming	Membership inactive
Telecom	Account terminated
Banking	No transactions for X months

Database Tables



Database: customer_churn

db_customer
customerid
name
country
State
gender
dob
interests
pincode



db_subscription
customerid
subscription_start_date
subscription_type
renewal_date
plan_type
contract_type
cancellation_date
cancellation_reason
monthly_charges
cltv
churn_score



db_support
customerid
complaint_date
Escalations
csat_score
col_1
comment

Calculated metrics

KPI	Formula
churn rate	churned customer / total customers
Churn by plan type	churn rate GROUP BY plan_type (Basic/Standard/Premium)
Churn by state	churn rate GROUP BY country, state
Retention rate	1-churn rate
ARPU	SUM(monthly_charges) / COUNT(active customerid)
Average customer tenure	AVG(DATEDIFF(cancellation_date OR NOW(), subscription_start_date))
Revenue at risk	SUM(monthly_charges) WHERE churn_score > 70
Escalation rate	SUM(escalations) / COUNT(complaints) × 100
Avg complaints per customer	COUNT(complaints) / COUNT(DISTINCT customerid)
Correlation: escalations → churn	churn rate WHERE escalations ≥ 1 vs 0

Insights

Insights

- Churn Rate: 28.6% | Retention Rate: 71.4%
- Most of the churn is from basic subscription plan – nothing to worry in terms of major revenue impact
- Most of the churn happened in the month of Sep 2024 and, most affected state is Karnataka
- Average Tenure (Days): 1,451 | ARPU is Rs 18.8
- Total Revenue = 395
- Revenue loss due to churn = 74 | CLTV Lost = 2,047
- % Revenue loss = 18%
- monthly vs annual churn = 55.6% vs 8.3%

Action Items

- Check what happened in Karnataka – was there any increase in price, user complaints, tech issues, etc
- Did we increased subs price for **Basic plan** or recently made any changes – specially in sep month?
- Check what competitor is doing – bcoz one user has moved to competitor
- Focus on customers with 'High' & 'Med' churn risk, check their LTV (to make priority list), complaint request, reach out to them through email, sms, calls and fix their issue.

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